

10.Network

This Section presents the network design produced for the accident-detection system, organized around the three elements required for evaluation: (1) the overall network architecture, (2) the important configuration settings that make the architecture function, and (3) the reasoning behind each major design decision. The design was fully built and validated in EVE-NG Community Edition using real vendor operating-system images, so every setting described below is empirically tested rather than theoretical.

10.1 Network Architecture

The architecture connects roadside cameras, switching fabric, compute servers, next-generation firewalls (NGFWs), and administrative users into a single resilient system. It is built as a multi-tier hierarchy — access, distribution/core, and security — with redundancy engineered into every tier so that no single link or device failure interrupts accident detection.

10.1.1 Global Topology

The topology below shows the full physical path from street-side cameras through the switching hierarchy to the compute servers, firewalls, and the outside world. Layer 3 redundancy (dual routing switches) and Layer 2 redundancy (ring links and dual core switches) are both shown.

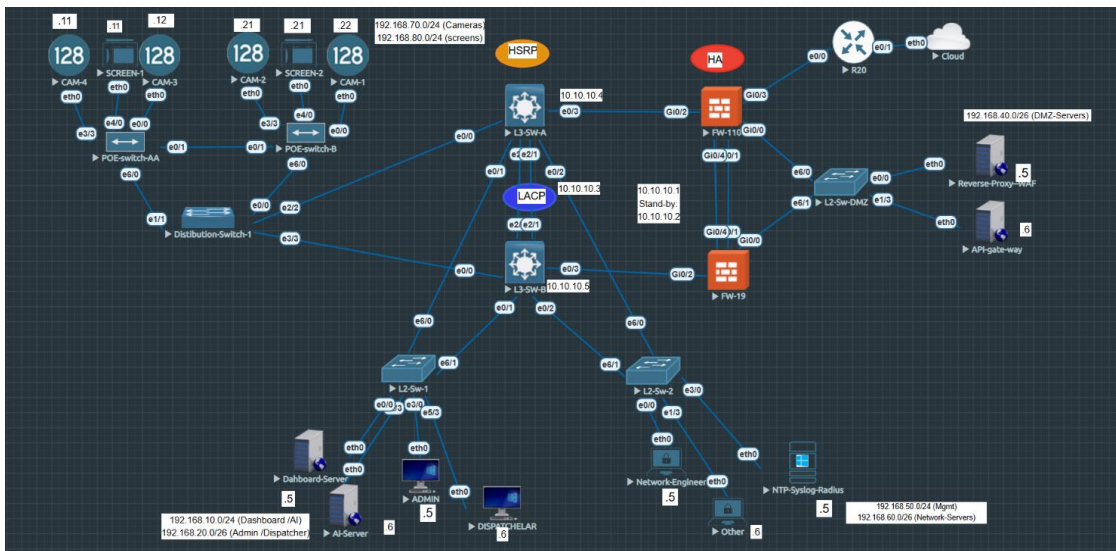


Figure 10.1 — Global network topology showing Layer 3 (HSRP/HA) redundancy.

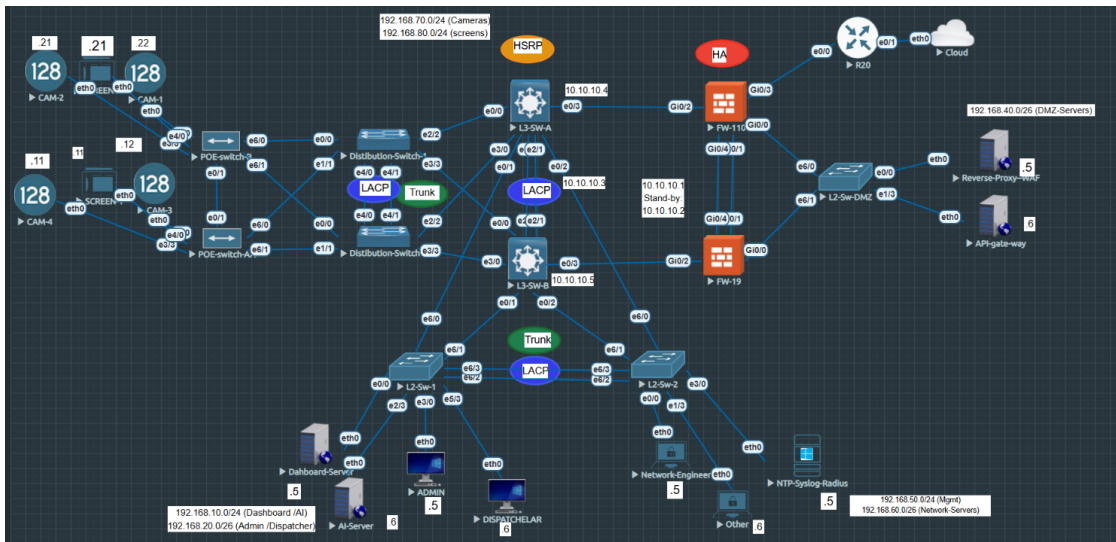


Figure 10.2 — Global network topology with added Layer 2 ring redundancy.

10.1.2 Functional Layered View

Logically, the network is organized into four functional layers: the Street/Access layer (cameras and screens), the Distribution/Core layer (switching and routing), the Security layer (firewall cluster and DMZ), and the Internal Services layer (AI, dashboard, and admin systems).

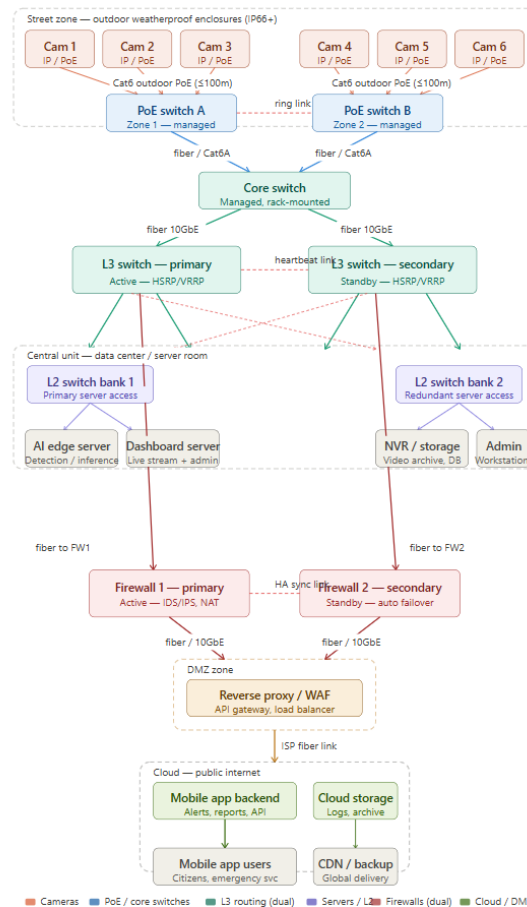


Figure 10.3 — Functional layered architecture diagram.

10.1.3 Physical Segments and Topology per Layer

Segment	Topology	Purpose
Camera → PoE switch	Star	Each camera has its own dedicated link
PoE switch ↔ PoE switch	Ring	Auto-failover if a street cable is cut
PoE switches → Core switch	Star uplinks	Both street zones uplink to the core
Core switch → dual L3 switches	Point-to-point ×2	One link to each L3 switch
L3 primary ↔ L3 secondary	Heartbeat link	HSRP keeps them in sync; secondary takes over in seconds
L3 switches → L2 switch banks	Cross-connected star	Each L3 switch reaches both L2 banks for full redundancy
L2 banks → servers	Star	Servers dual-homed to both L2 banks where possible
L3 switches → dual firewalls	Point-to-point ×2	Each L3 switch feeds one firewall
FW1 ↔ FW2	HA sync link	Active/standby — FW2 takes over instantly if FW1 fails
Firewalls → DMZ	Converged	Both firewalls feed the same DMZ zone
Firewalls → Cloud	Single ISP link	A second ISP link can be added for full WAN redundancy

Table 10.1 — Hybrid topology: pathing and redundancy mechanism per network segment.

10.1.4 Core Hardware (For Real Implementation (not in emulation))

Device	Model	Qty	Role
Outdoor PoE switch	Cisco Catalyst IE9320-24P4S	2	Street camera power + data
Core switch	Cisco Catalyst 9500-24Y4C	1–2	Aggregation backbone
L3 routing switch	Cisco Catalyst 9300X-48HX	2	Inter-VLAN routing, HSRP
L2 access switch	Cisco Catalyst 9300-48P	2–4	Server access layer
Firewall / NGFW	Fortinet FortiGate 200G	2	Security, HA active/standby

Table 10.2 — Core hardware inventory and functional role of each device.

- **Outdoor PoE switch** — carries power and data to roadside cameras over one Ethernet cable, cutting installation and cabling cost.
- **Core switch** — aggregates all access-switch traffic and forwards it to the routing layer.
- **L3 routing switch** — performs inter-VLAN routing and HSRP gateway redundancy between network segments.
- **L2 access switch** — connects servers and internal devices with VLAN segmentation.
- **Firewall / NGFW** — filters traffic, enforces policy, and isolates public-facing services in the DMZ.

10.1.5 VLAN Architecture

Traffic is segmented logically into seven VLANs so that camera, server, admin, management, and internet-facing traffic never share a broadcast domain, even though they run over the same physical switches.

VLAN	Name	Zone	Purpose
10	Dashboard / AI	Internal	Receives camera streams, runs inference, pushes results to admin dashboard
20	Admin / Dispatcher	Internal	Monitors live streams and dispatches alerts
40	DMZ	DMZ	Internet-facing services buffer zone (reverse proxy, WAF, API gateway)
50	IT-Mgmt	Internal	Out-of-band device management
60	Network-Servers	Internal	DHCP, syslog, NTP
70	Cameras	Street	Isolated camera traffic — no inter-VLAN access except to AI servers
80	Screen	Street	Displays open/closed lane status

Table 10.3 — VLAN segmentation and security zoning.

10.1.6 Cabling Overview (For Real Implementation (not in emulation))

Media selection follows the demand of each segment: outdoor runs use shielded, UV-rated copper or single-mode fiber for interference resistance and distance, while short intra-rack links use Cat6A or DAC for cost efficiency.

Segment	Cable	Reason
Camera → PoE switch	Cat6 outdoor STP, direct-burial, UV-rated	Carries PoE power + data; withstands weather
PoE switch ↔ PoE switch (ring)	Cat6A shielded / single-mode fiber	Fiber used when the run exceeds 100 m or nears EMI sources
PoE switches → Core switch	Single-mode fiber 10G (SFP+)	High-bandwidth, interference-free outdoor backbone
Core → L3 switches	Fiber or DAC copper, 10/25G	Short rack runs, line-rate throughput
L3 ↔ L3 heartbeat	Cat6A, short dedicated run	Isolated HSRP heartbeat traffic
L3 → L2 switch banks	Cat6A / OS2 fiber SFP+	Distribution-to-access connectivity
L2 → servers	Cat6A, dual NIC	Standard 1/10GbE with NIC redundancy
Firewalls ↔ ISP	OS2 single-mode fiber	ISP-provided high-bandwidth handoff

Table 10.4 — Physical media selected per segment.

10.2. Important Configuration Settings

This section documents the specific settings that were configured to make the architecture above actually work — addressing, gateway redundancy, VLAN tagging, routing, and firewall policy. All values were applied and verified in the EVE-NG emulation.

10.2.1 IP Addressing Plan

The network uses RFC 1918 private addressing internally under the 192.168.0.0/16 block, with a fixed hierarchical scheme so that ACLs, monitoring, and future growth stay predictable.

- **Camera subnet policy** — all camera interfaces are DHCP-reserved by MAC address so they retain the same IP after reboot.
- **Server subnet policy** — infrastructure servers use static IPs assigned directly on the interface.
- **Network isolation** — the Quarantine network hands out leases with no default gateway or DNS to isolate unauthenticated devices.
- **Gateway rule** — the .1 address in every subnet is reserved for the HSRP virtual gateway and is never assigned to physical hardware.
- **Scale projection** — the camera subnet uses a /24 mask specifically to leave headroom for future zone expansion.

VLAN	Name	Subnet	Gateway	Hosts
70	Cameras	192.168.70.0/24	192.168.70.1	253
80	Screen	192.168.80.0/24	192.168.80.1	253
10	Dashboard / AI / Storage	192.168.10.0/24	192.168.10.1	253
20	Admin / Dispatcher	192.168.20.0/26	192.168.20.1	61
50	Mgmt	192.168.50.0/24	192.168.50.1	253
60	Network-Servers	192.168.60.0/26	192.168.60.1	61
40	DMZ	192.168.40.0/26	192.168.40.1	61

Table 10.5 — Subnet allocation under the private 192.168.0.0/16 block.

10.2.2 HSRP Gateway Redundancy Settings

Every internal VLAN gateway is a shared HSRP virtual IP spread across the two Layer 3 switches, so a switch failure does not take down the default gateway for that segment.

VLAN	Virtual Gateway IP	L3 Primary IP	L3 Secondary IP
10	192.168.10.1	192.168.10.2	192.168.10.3
20	192.168.20.1	192.168.20.2	192.168.20.3
40	192.168.40.1	192.168.40.2	192.168.40.3
50	192.168.50.1	192.168.50.2	192.168.50.3

VLAN	Virtual Gateway IP	L3 Primary IP	L3 Secondary IP
60	192.168.60.1	192.168.60.2	192.168.60.3
70	192.168.70.1	192.168.70.2	192.168.70.3
80	192.168.80.1	192.168.80.2	192.168.80.3

Table 10.6 — HSRP virtual gateway mapping per VLAN.

Failover rule: the primary switch runs HSRP priority 110 with preemption enabled; the secondary runs priority 100. If the primary goes down, the secondary assumes the gateway role within roughly 3 seconds.

```
L3-SW-A#show standby brief
                P indicates configured to preempt.
                |
Interface    Grp  Pri  P State   Active        Standby        Virtual IP
Et0/3        1   110  P Active  local         unknown        10.10.10.3
Vl10         10  110  P Active  local         192.168.10.3   192.168.10.1
Vl20         20  110  P Active  local         192.168.20.3   192.168.20.1
Vl50         50  110  P Active  local         192.168.50.3   192.168.50.1
Vl60         60  110  P Active  local         192.168.60.3   192.168.60.1
Vl70         70  110  P Active  local         192.168.70.3   192.168.70.1
Vl80         80  110  P Active  local         192.168.80.3   192.168.80.1
```

Fig(10.4): Active Verification Status log displaying working HSRP configuration mapping.

```
L3-SW-B
*Jul  3 15:11:35.193: %HSRP-5-STATECHANGE: Vlan80 Grp 80 state Speak -> Standby
L3-SW-B(config)#
*Jul  3 15:11:35.710: %HSRP-5-STATECHANGE: Vlan50 Grp 50 state Speak -> Standby
*Jul  3 15:11:35.925: %HSRP-5-STATECHANGE: Vlan20 Grp 20 state Speak -> Standby
*Jul  3 15:11:36.601: %HSRP-5-STATECHANGE: Vlan60 Grp 60 state Speak -> Standby
L3-SW-B(config)#exit
L3-SW-B#wr
*Jul  3 15:12:17.672: %SYS-5-CONFIG_I: Configured from console by console
L3-SW-B#write
Building configuration...
Compressed configuration from 3076 bytes to 1543 bytes[OK]
L3-SW-B#show running-config interface Port-channel1
Building configuration...

Current configuration : 177 bytes
!
interface Port-channel1
 description Direct-link-to-L3-SW-A
 switchport trunk allowed vlan 10,20,50,60,70,80
 switchport trunk encapsulation dot1q
 switchport mode trunk
end
L3-SW-B#
```

Fig (10.5)— HSRP Standby State Confirmation and Port-Channel1 Trunk Configuration on L3-SW-B

10.2.3 VLAN Tagging and Trunking Settings

- **Trunk standard** — inter-switch trunks use IEEE 802.1Q with an explicit allowed-VLAN list on every trunk.
- **Native VLAN** — all trunks use an isolated, non-routed VLAN 99 with no active hosts, eliminating double-tagging attacks.
- **Edge access hardening** — camera-facing switch ports are hard-set as access ports in VLAN 70 with dynamic trunk negotiation disabled.
- **Stateful failover separation** — HA/heartbeat traffic (VLAN 100, VLAN 110) runs on dedicated physical interconnects, excluded from the general switch trunks.

```
L3-SW-A#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Et1/0, Et1/1, Et1/2, Et1/3 Et2/0, Et2/1, Et2/2, Et2/3 Et3/0, Et3/1, Et3/2, Et3/3
10	Dash/AI	active	
20	Admin/Disp	active	
50	IT-mgt	active	
60	network-serv	active	
70	Cameras	active	
80	Screen	active	
1002	fddi-default	act/unsup	
1003	token-ring-default	act/unsup	
1004	fddinet-default	act/unsup	
1005	trnet-default	act/unsup	

```
L3-SW-A#
```

Fig(10.6): VLAN Allocation verification on Primary Distribution Switch (L3-SW-A)

```
L3-SW-A#show interfaces trunk
```

Port	Mode	Encapsulation	Status	Native vlan
Et0/0	on	802.1q	trunking	1
Et0/1	on	802.1q	trunking	1
Et0/2	on	802.1q	trunking	1

Port	Vlans allowed on trunk
Et0/0	50,70,80
Et0/1	10,20,50
Et0/2	50,60

Port	Vlans allowed and active in management domain
Et0/0	50,70,80
Et0/1	10,20,50
Et0/2	50,60

Port	Vlans in spanning tree forwarding state and not pruned
Et0/0	50,70,80
Et0/1	10,20,50
Et0/2	50,60

Fig(10.7): Operational Trunking Configuration Output

10.2.4 Routing Protocol Settings

Protocol	Where Used	Setting / Behavior
OSPF	Between the dual L3 switches and the core switch	Dynamic link-state routing; reconverges automatically within seconds of a link failure
Static routing	L3 switches → firewalls → DMZ	Fixed routes on paths that never change, avoiding routing-protocol overhead
Inter-VLAN routing	L3 switch SVIs	Routes camera, server, and admin VLANs to each other without an external router
NAT	Firewalls	Translates internal private addresses to public IPs, hiding internal topology

Table 10.7 — Routing protocol configuration and behavior.

```

L3-SW-A#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.10.10.0/29 is directly connected, Ethernet0/3
L       10.10.10.4/32 is directly connected, Ethernet0/3
192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.10.0/24 is directly connected, Vlan10
L       192.168.10.2/32 is directly connected, Vlan10
192.168.20.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.20.0/26 is directly connected, Vlan20
L       192.168.20.2/32 is directly connected, Vlan20
192.168.40.0/26 is subnetted, 1 subnets
S       192.168.40.0 [1/0] via 10.10.10.1
192.168.50.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.50.0/24 is directly connected, Vlan50
L       192.168.50.2/32 is directly connected, Vlan50
192.168.60.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.60.0/26 is directly connected, Vlan60
L       192.168.60.2/32 is directly connected, Vlan60
192.168.70.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.70.0/24 is directly connected, Vlan70
L       192.168.70.2/32 is directly connected, Vlan70
192.168.80.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.80.0/24 is directly connected, Vlan80
L       192.168.80.2/32 is directly connected, Vlan80

```

Fig(10.8): Main Routing Table on Layer 3 Switch Alpha (L3-SW-A), proving functional Inter-VLAN resolution.

```
L3-SW-A#show ip ospf neighbor

Neighbor ID    Pri   State           Dead Time   Address        Interface
1.1.2.2        1     FULL/DR         00:00:35   192.168.10.3   Vlan10
1.1.3.3        1     FULL/BDR        00:00:35   10.10.10.1     Ethernet0/3
L3-SW-A#
```

Fig(10.9): OSPF Adjacency table on L3-SW-A, establishing dynamic links between redundant distributions.

```
FW1# show route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, + - replicated route
Gateway of last resort is not set

C      10.10.10.0 255.255.255.248 is directly connected, inside
L      10.10.10.1 255.255.255.255 is directly connected, inside
C      172.16.255.0 255.255.255.252 is directly connected, FO
L      172.16.255.1 255.255.255.255 is directly connected, FO
C      172.16.255.4 255.255.255.252 is directly connected, STATEFUL
L      172.16.255.5 255.255.255.255 is directly connected, STATEFUL
S      192.168.10.0 255.255.255.0 [1/0] via 10.10.10.3, inside
C      192.168.40.0 255.255.255.192 is directly connected, dmz
L      192.168.40.1 255.255.255.255 is directly connected, dmz
```

Fig(10.10): Core Security Appliance (FW1) Routing Table, outlining explicit pathing strategies for internal and DMZ traffic blocks.

10.2.5 Firewall Security Zones and ACL Settings

Zone	Contains	Trust Level
Internal	AI server, dashboard, NVR, servers	Trusted
DMZ	Reverse proxy, WAF, API gateway	Semi-trusted
External / Internet	Mobile app users, cloud, ISP	Untrusted

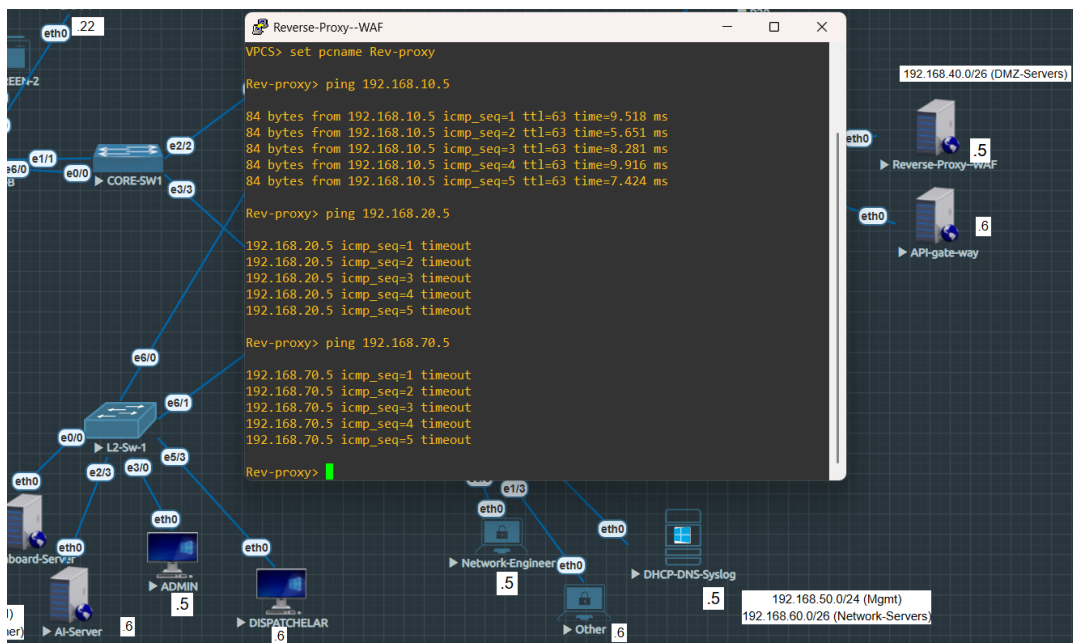
Table 10.8 — Firewall security zone definitions.

The firewall enforces least-privilege, default-deny policy. Key rules include:

- camera streams (TCP/UDP) are allowed only from the Street zone into the AI/Dashboard servers.
- the AI server may initiate RTSP sessions back to cameras, and pushes detection results to the dashboard.
- Admin/Dispatcher stations may reach internal servers for monitoring, and send instructions back.
- internal servers may respond to the DMZ reverse proxy, but the DMZ cannot initiate sessions inward.
- any traffic not explicitly permitted is denied and logged by default.

```
FW1# show access-list
access-list cached ACL log flows: total 0, denied 0 (deny-flow-max 4096)
      alert-interval 300
access-list INSIDE_TO_DMZ; 1 elements; name hash: 0xb3428d0e
access-list INSIDE_TO_DMZ line 1 extended permit ip 192.168.10.0 255.255.255.0 1
192.168.40.0 255.255.255.192 (hitcnt=28) 0xb14f6f51
access-list DMZ_TO_INSIDE; 1 elements; name hash: 0xed3b440d
access-list DMZ_TO_INSIDE line 1 extended permit ip 192.168.40.0 255.255.255.192
192.168.10.0 255.255.255.0 (hitcnt=0) 0x0d65bedf
access-list DMZ_TO_OUTSIDE; 1 elements; name hash: 0x90e99f49
access-list DMZ_TO_OUTSIDE line 1 extended permit ip 192.168.40.0 255.255.255.19
2 any (hitcnt=30) 0x4fb0e2e9
```

Fig (10.11): Live Policy access configuration metrics pulled from Firewall 1 (FW1).



Fig(10.12): ICMP Proof confirming isolation constraints; DMZ hosts are dropped when attempting to touch internal zones outside of core AI paths.

10.2.6 Firewall High-Availability Settings

Link	Traffic Type	Interval / Behavior
HA-1 Heartbeat	UDP hello packets	Sent every 1 second; 15 missed hellos triggers failover to standby
HA-2 Session sync	TCP, stateful	Every active session/NAT table entry mirrored to standby in real time
HA-3 Config sync	TCP	Every configuration change is auto-pushed to the standby unit

Table 10.9 — Firewall HA synchronization link settings.

```
FW-110
FW1(config)# int g0/1
FW1(config-if)# no sh
FW1(config-if)# int g0/4
FW1(config-if)# no sh
FW1(config-if)# exit
FW1(config)# failover lan unit primary
FW1(config)# failover lan interface FO g0/1
INFO: Non-failover interface config is cleared on GigabitEthernet0/1 and its sub
FW1(config)# failover interface ip FO 172.16.255.1 255.255.255.252 standby 1725
FW1(config)# failover link STATEFUL g0/4
INFO: Non-failover interface config is cleared on GigabitEthernet0/4 and its sub
FW1(config)# failover interface ip STATEFUL 172.16.255.5 255.255.255.252 stand5
FW1(config)# failover
FW1(config)# EXIT
FW1#
FW1# write
Building configuration...
Cryptochecksum: cf557ab1 20c8238d cdb967d8 29704c08
6874 bytes copied in 0.330 secs
[OK]
FW1#
Warning: ASAv platform license state is Unlicensed.
Install ASAv platform license for full functionality.
.
No Active mate detected
Beginning configuration replication: Sending to mate.
End Configuration Replication to mate
FW1# show failover
Failover On
Failover unit Primary
Failover LAN Interface: FO GigabitEthernet0/1 (up)
Reconnect timeout 0:00:00
Unit Poll frequency 1 seconds, holdtime 15 seconds
Interface Poll frequency 5 seconds, holdtime 25 seconds
Interface Policy 1
Monitored Interfaces 0 of 61 maximum
MAC Address Move Notification Interval not set
Version: Ours 9.6(1)S, Mate 9.6(1)S
Serial Number: Ours 9A27EB7GLMH, Mate 9A6XRQBD7JM
Last Failover at: 23:46:16 UTC Apr 19 2026
This host: Primary - Active
Active time: 56 (sec)
slot 0: empty
Other host: Secondary - Standby Ready
Active time: 0 (sec)

FW-19
FW1(config)# failover lan unit secondary
FW2(config)# failover lan interface FO g0/1
INFO: Non-failover interface config is cleared on GigabitEthernet0/1 and its sub
-interfaces
FW2(config)# failover interface ip FO 172.16.255.1 255.255.255.252 standby 1725
FW2(config)# failover link STATEFUL g0/4
INFO: Non-failover interface config is cleared on GigabitEthernet0/4 and its sub
-interfaces
FW2(config)# failover interface ip STATEFUL 172.16.255.5 255.255.255.252 stand5
FW2(config)# failover
FW2(config)# e.
.
Detected an Active mate
Beginning configuration replication from mate.
WARNING: Disabling auto import may affect Smart Licensing
Creating trustpoint "SmartCallHome_ServerCA" and installing certificate...
Trustpoint CA certificate accepted.
WARNING: Trustpoint SmartCallHome_ServerCA is already authenticated.
ERROR: % Ambiguous command: "e"
FW1(config)# End configuration replication from mate.
FW1(config)# exit
FW1# write
Building configuration...
Cryptochecksum: 86d292cf 4f153967 ebd0912 5499bafa
6876 bytes copied in 0.330 secs
[OK]
FW1# show failover
Failover On
Failover unit Secondary
Failover LAN Interface: FO GigabitEthernet0/1 (up)
Reconnect timeout 0:00:00
Unit Poll frequency 1 seconds, holdtime 15 seconds
Interface Poll frequency 5 seconds, holdtime 25 seconds
Interface Policy 1
Monitored Interfaces 0 of 61 maximum
MAC Address Move Notification Interval not set
Version: Ours 9.6(1)S, Mate 9.6(1)S
Serial Number: Ours 9A6XRQBD7JM, Mate 9A27EB7GLMH
Last Failover at: 23:45:12 UTC Apr 19 2026
This host: Secondary - Standby Ready
Active time: 0 (sec)
slot 0: empty
Other host: Primary - Active
Active time: 59 (sec)
```

Fig(10.13): Dual Console verification layout showing active state replication from Firewall 1 over to the standby node.

10.3. Reasons for Choosing This Design

Each major decision below was evaluated against the operating requirement of the system: detecting road accidents and alerting responders within a strict, safety-critical time budget.

10.3.1 Why Wired Infrastructure Was Chosen

Feature	Wired	Wireless
Speed	High	Medium
Stability	Very stable	Affected by interference
Security	More secure	More vulnerable
Mobility	Low	High
Cost	More expensive upfront	Cheaper initially, may cost more long-term at scale

Table 10.10 — Wired vs. wireless comparison for the edge environment.

Wired infrastructure was selected because the system is safety-critical and cannot tolerate the variability of radio links:

- higher throughput for servers and routing.
- stable and reliable, with no interference — critical for real-time accident detection.
- lower latency, giving faster AI inference response when every second counts.
- higher bandwidth to support multiple 4K camera streams simultaneously.
- more secure — much harder to intercept or jam than wireless.
- Power over Ethernet lets a single cable carry both data and power, simplifying roadside installation.

10.3.2 Why Local Edge Processing Was Chosen Over Cloud

Feature	Local / Edge	Cloud
Processing location	AI runs on-site	Video sent to cloud for processing
Latency	Ultra-low, real-time	Higher, due to internet round-trip
Internet dependency	Works offline	Requires stable, high-bandwidth internet
Data privacy	High — video stays on-premises	Lower — data leaves the site
Reliability	Independent of internet outages	May stop if internet goes down

Table 10.11 — Local edge vs. cloud processing trade-offs.

For accident detection, every millisecond matters, and the system must keep working even if the internet link fails. Running inference on a local GPU server (fed by roadside cameras) removes the cloud round-trip entirely and keeps detection functioning during an outage — the two properties a safety system cannot compromise on.

10.3.3 Why This Redundancy Model Was Chosen

Protocol / Mechanism	Where Used	Why It Was Chosen
HSRP	Dual L3 switches	One shared virtual gateway IP; secondary takes over in under 3 seconds if the primary fails
Firewall Active/Standby HA	Dual FortiGate 200G	Session state is mirrored continuously, so failover is transparent and connections survive
LACP	L3 switch uplinks	Bundles physical ports into one logical link — failover with zero latency penalty and more bandwidth
Rapid-PVST+	Ring links, dual core switches	Prevents Layer 2 loops from the redundant paths while keeping a backup link ready
Dual STP-arbitrated uplinks	PoE/L2 switches to dual core/L3	Used where LACP cannot apply because each uplink terminates on a different physical switch
Redundant core switch pair	Core-SW-1 / Core-SW-2	Removes the single core switch as a point of failure for all camera and screen traffic

Table 10.12— Redundancy mechanisms and the reasoning behind each.

The guiding principle is that redundancy should cost nothing during normal operation and only introduce a brief, one-time delay at the moment of an actual failure — which is an acceptable trade-off for a system required to stay online continuously.

10.3.4 Why the Latency Budget Drove Key Decisions

The design targets under 200 ms from camera capture to dashboard alert, and under 1500 ms to a mobile push notification. This single requirement justifies several choices made elsewhere in this report:

- wired-only links, because Wi-Fi's 5–50 ms of variable latency and packet loss would consume too much of the budget.
- local edge inference, because a cloud round-trip alone could exceed the entire dashboard-alert budget.
- fiber for all inter-switch links, keeping per-hop switching latency under 1 ms.
- hardware-ASIC firewalls (FortiGate 200G), keeping deep packet inspection latency under 5 ms even under load.

Network switching and routing together are budgeted to contribute under 5 ms combined, leaving the AI inference stage (20–50 ms) as the dominant — and expected — contributor to total latency.

10.3.5 Why This Cost Profile Was Accepted

Networking hardware (roughly \$110,000) and cabling/physical infrastructure (roughly \$7,300) bring total hardware capital expenditure to approximately \$117,300. This upfront cost was accepted,

rather than a lower-cost cloud/wireless alternative, because the recurring savings of no cloud fees, the elimination of internet-dependency risk, and the latency guarantees of a wired local-edge design directly serve the safety-critical purpose of the system — a trade-off that only makes sense once the sections above are considered together.